

Implementation of Logistic Regression (LR) in Python

Aim

To implement the **Logistic Regression (LR)** algorithm in Python for classification and evaluate its performance using **Accuracy** and a **Confusion Matrix**.

Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
 - NumPy
 - Pandas
 - Scikit-learn (sklearn)

Theory

Logistic Regression is a **supervised machine learning algorithm** used for **classification** problems. It predicts the probability that a given input belongs to a particular class using the **sigmoid (logistic) function**. It is widely used for binary and multi-class classification tasks.

Advantages:

- Easy to implement and interpret.
- Fast and efficient.
- Works well for linearly separable data.

Disadvantages:

- Not suitable for complex non-linear datasets.
- Sensitive to outliers.

Algorithm

1. Import the required libraries.

2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Standardize the feature values.
5. Train the Logistic Regression model.
6. Predict the class labels for the test data.
7. Calculate the Accuracy Score.
8. Display the Confusion Matrix.

Code (Google Colab)

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Feature Scaling
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Logistic Regression Model
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Results
print("Predicted Classes:")
print(y_pred)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))
```

Sample Output

```
... Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

Accuracy:
0.9777777777777777
```

Result

The **Logistic Regression (LR)** algorithm was successfully implemented in Python using the Iris dataset. The classifier predicted the test samples with high accuracy, and its performance was evaluated using the **Confusion Matrix** and **Accuracy Score**.